

IDRINSKOYE, Russian Federation

WMO: 297660

Lat: 54.365N

Lon: 92.150E

Elev: 282

StdP: 97.98

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.4	(c) -35.1	(d) -40.7	(e) 0.1	(f) -38.3	(g) -37.5	(h) 0.1	(i) -35.0	(j) 6.1	(k) -4.6	(l) 5.2	(m) -4.6	(n) 0.7	(o) 110	(p) 0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	30.4	19.7	28.6	18.9	26.7	18.2	21.0	27.9	20.0	26.4	19.0	25.1	1.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.0	23.7	17.7	13.1	22.6	16.8	12.4	21.8	62.4	28.0	58.7	26.3	55.5	25.1	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.6	(b) 4.7	(c) 4.0	(e) -38.9	(f) 33.9	(g) 4.1	(h) 2.1	(i) -41.8	(j) 35.4	(k) -44.2	(l) 36.6	(m) -46.5	(n) 37.8	(o) -49.4	(p) 39.3		
(a) 5.6	(b) 4.7	(c) 4.0	(e) -38.9	(f) 33.9	(g) 4.1	(h) 2.1	(i) -41.8	(j) 35.4	(k) -44.2	(l) 36.6	(m) -46.5	(n) 37.8	(o) -49.4	(p) 39.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.1	-19.5	-16.5	-6.7	3.4	10.3	17.2	19.1	16.5	9.6	2.3	-7.4	-15.5		
	DBStd	14.70	9.24	8.57	7.36	5.07	4.83	3.82	2.82	3.50	4.14	4.78	7.45	8.66		
	HDD10.0	4051	916	742	518	203	57	2	0	1	57	242	523	792		
	HDD18.3	6376	1174	975	777	448	252	64	25	77	263	498	773	1050		
	CDD10.0	821	0	0	0	6	66	218	281	204	45	2	0	0		
	CDD18.3	103	0	0	0	0	3	31	48	21	1	0	0	0		
	CDH23.3	1364	0	0	0	2	78	461	523	272	29	0	0	0		
	CDH26.7	384	0	0	0	0	17	141	148	71	7	0	0	0		
	WSAvg	1.6	1.1	1.2	1.5	2.0	2.1	1.7	1.5	1.5	1.6	1.7	1.6	1.3		
	PrecAvg	384	13	12	9	18	32	56	62	66	45	28	23	20		
(15) Precipitation	PrecMax	474	25	36	25	37	84	93	117	163	107	75	39	47		
	PrecMin	268	2	3	1	1	10	3	24	29	11	9	6	6		
	PrecStd	51	6	8	6	9	17	22	23	32	19	13	9	10		
	PrecStd	51	6	8	6	9	17	22	23	32	19	13	9	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	2.3	10.8	22.2	29.2	32.6	32.3	31.3	27.0	17.9	9.4	1.9		
		MCWB	0.6	-0.3	5.1	11.7	16.2	19.2	20.9	20.5	16.7	10.6	6.0	-0.8		
	2%	DB	-0.8	0.2	7.2	18.3	25.8	30.2	30.3	28.7	22.4	13.8	6.4	-0.2		
		MCWB	-2.6	-1.7	3.0	10.0	15.0	19.0	20.3	19.5	15.2	8.6	3.2	-2.1		
	5%	DB	-3.5	-1.7	5.2	15.3	23.2	28.2	28.5	26.4	19.6	11.3	3.9	-2.1		
		MCWB	-5.0	-3.4	1.9	8.6	13.6	18.1	19.6	18.6	13.5	7.5	1.7	-3.8		
	10%	DB	-5.9	-3.9	3.3	12.4	20.5	26.1	26.4	24.2	17.1	9.3	2.0	-3.9		
		MCWB	-7.2	-5.3	0.6	6.9	12.5	17.4	18.9	17.5	12.2	6.3	0.4	-5.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	0.2	5.2	13.0	17.7	21.2	22.8	21.9	17.6	11.2	6.3	-0.3		
		MCDB	2.8	1.8	9.7	21.9	27.1	29.4	29.1	28.8	25.0	15.7	9.1	1.3		
	2%	WB	-2.4	-1.5	3.3	10.7	15.9	20.0	21.4	20.4	15.6	9.3	3.5	-2.0		
		MCDB	-0.9	0.2	6.7	17.3	23.6	27.5	28.1	27.0	21.3	12.9	5.8	-0.3		
	5%	WB	-5.0	-3.3	2.0	8.9	14.5	19.0	20.4	19.2	14.3	7.9	1.9	-3.7		
		MCDB	-3.6	-1.7	4.9	14.6	21.6	26.5	26.6	25.1	18.6	11.1	3.6	-2.3		
	10%	WB	-7.2	-5.3	0.7	7.3	13.1	18.0	19.5	18.0	12.8	6.5	0.4	-5.0		
		MCDB	-6.0	-3.8	3.2	12.2	19.3	24.9	25.1	23.0	16.2	9.0	2.0	-3.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	11.0	11.9	11.5	12.9	12.9	11.6	11.6	11.3	9.3	7.8	8.5		
		MCDBR	8.0	9.5	12.1	17.5	18.4	17.7	15.7	15.9	16.3	13.3	7.7	7.6		
		MCWBR	7.0	7.9	8.4	10.2	9.1	7.6	6.8	7.4	8.7	8.3	5.6	6.5		
		MCDBR	8.0	9.1	11.2	15.9	15.8	15.0	13.3	14.0	14.5	12.5	7.2	7.5		
	5% WB	MCWBR	7.0	7.7	7.9	9.7	8.3	6.7	6.3	6.8	7.9	8.0	5.3	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.322	0.370	0.412	0.422	0.410	0.423	0.411	0.372	0.354	0.325	0.291		
	taud		2.026	1.985	1.927	2.066	2.155	2.296	2.280	2.283	2.325	2.234	2.056	1.969		
	Ebn at Noon		621	727	776	806	823	839	818	804	788	698	557	506		
	Edh at Noon		72	107	143	144	139	123	123	115	97	84	71	62		
(44) All-Sky Solar Radiation	RadAvg		0.96	1.98	3.37	4.35	5.07	5.96	5.31	4.43	3.04	1.66	0.84	0.62		
	RadStd		0.10	0.14	0.32	0.25	0.58	0.58	0.38	0.38	0.35	0.16	0.11	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page